

Who Checks the Checker? Auditing Automated Demonstration Scoring for Teleoperated Robot Data, and a Telemetry-First Verifier Stack

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Abstract

Deciding whether a robot completed a task underlies training-data curation, policy evaluation, reinforcement-learning reward and the decision to hand control to a remote operator. Automated scoring services now offer such verdicts for teleoperated demonstrations, but their reliability is rarely measured. We audit one commercial service on 323 teleoperated bimanual episodes (9.4 hours) from a public humanoid teleoperation sample. Headline scores are high (mean 8.88 of 10; 84.2% success) yet fail several reliability checks: scores are derived entirely from a single camera view with zero weight on robot telemetry; human corrections only raise scores; 70% of episodes are flagged for reviewer disagreement; about 1% of episodes receive a *clean* quality grade despite 90% scoring 8 or higher; and score is uncorrelated with episode duration (Pearson $r = 0.07$). Operator-level scoring could not run because the data carried no operator identifiers. We conclude that video-only scoring is suitable for triage but not for filtering training data. We then propose a telemetry-first verifier ensemble with two-way correction, disagreement routing to calibrated humans and an auditable evidence ledger that serves data quality control, policy evaluation and runtime operations from a single stack, together with a validation protocol that accepts a scoring method only if it predicts downstream policy performance.

1 Introduction

Physical-AI systems are increasingly trained on teleoperated demonstrations and deployed with a remote human in the loop. Every stage of this pipeline relies on a *verifier*: a procedure that decides whether an attempted task succeeded and how well it was executed. Verifiers select training data, compare policies, supply reward for reinforcement learning, and decide at run time when a robot is stuck and an operator must intervene. A recent survey of roughly 150 verifiers for robot policies argues that verification faces an inherent trade-off: credibility falls as availability rises [1]. Human experts are credible but slow and costly; automated checks are cheap and constant but less trustworthy.

Commercial services have begun to sell automated demonstration scoring to robotics teams [8, 10]. If such scores are used to filter training data or to certify datasets for buyers, their failure modes propagate directly into trained policies and into commercial claims. Yet independent audits of these scores are scarce. This paper makes three contributions:

- **An audit** of a commercial demonstration-scoring service on 323 public teleoperation episodes, quantifying score distribution, label consistency, reviewer agreement and sensitivity to camera choice (Sections 3 and 4).

- **A design** for a telemetry-first verifier stack and evidence ledger that serves data curation, policy evaluation and runtime operations from a single set of verdicts (Section 5).
- **A validation protocol** that accepts a scoring method only if it predicts downstream policy performance, agrees with expert humans without directional bias, and is invariant to camera choice (Section 6).

2 Related Work

Verifiers for robot policies. Wan et al. [1] categorize verifiers by judgment source (human, rule-based or formal, learned, and model-intrinsic) and document the credibility-availability trade-off and the risk of reward hacking when verifiers are used for training. Our audit is an empirical instance of that trade-off for a learned, video-based verifier.

Policy evaluation infrastructure. RoboArena [2] scales real-world evaluation through double-blind pairwise comparisons across institutions. XPolicyLab [3] standardizes the interface between policies and simulated or real evaluation environments. Open simulation benchmarks [7] provide reproducible evaluation in simulation. These systems address *where* policies are evaluated; we focus on *how* success is judged, which all of them depend on.

Data formats and multimodal signals. LeRobot [4] standardizes robot-learning datasets, including tabular state and action streams alongside video. Tactile-aware VLAs such as T-Rex [5] show that touch substantially improves force-sensitive manipulation, implying that verifiers which ignore non-visual channels miss the signals most relevant to contact-rich tasks. DUET [6] shows that human-human demonstrations can substitute for much robot teleoperation, increasing the importance of judging the quality of whatever robot data remains.

Data-quality products. Commercial services now score teleoperation, handheld-gripper and egocentric demonstrations per operator, task and episode [8], or offer data-readiness checks and annotation layers [10]. Robotics visualization platforms support inspection but do not issue verdicts.

3 Audit Setup

3.1 Data

We use a publicly released, access-gated teleoperation sample of about ten hours [9], recorded on a wheeled bimanual humanoid and stored in the LeRobot v3.0 format [4] at 30 fps. Each frame carries a 52-dimensional action, a 290-dimensional state vector, a flag indicating the action source, and five camera streams (Figure 1). Tasks include RAM-module insertion, refilling and extinguishing alcohol lamps, packing medicine into first-aid kits, packing and taping boxes, wiping lockers and handling documents.

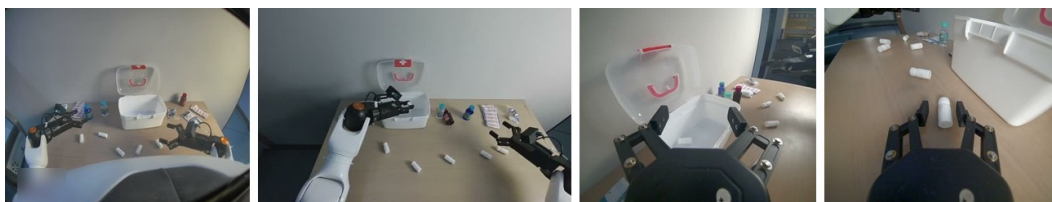


Figure 1: Four of the five camera streams for the task *place scattered medicine bottles into the first aid box* (left to right: head stereo, top head, left wrist, right wrist). Frames are drawn from different episodes of the same task.

3.2 Scoring system under audit

The episodes were indexed by a commercial demonstration-scoring service, hereafter *Service A* [8], in a workspace we control. The service assigns each episode a quality score out of 10 (a weighted blend of facet scores such as arm visibility, task engagement, object interaction and task completion), an analysis outcome label, a review quality grade (clean, mixed, rough or needs review), and free-text evidence with reviewer dissent (Figure 2).

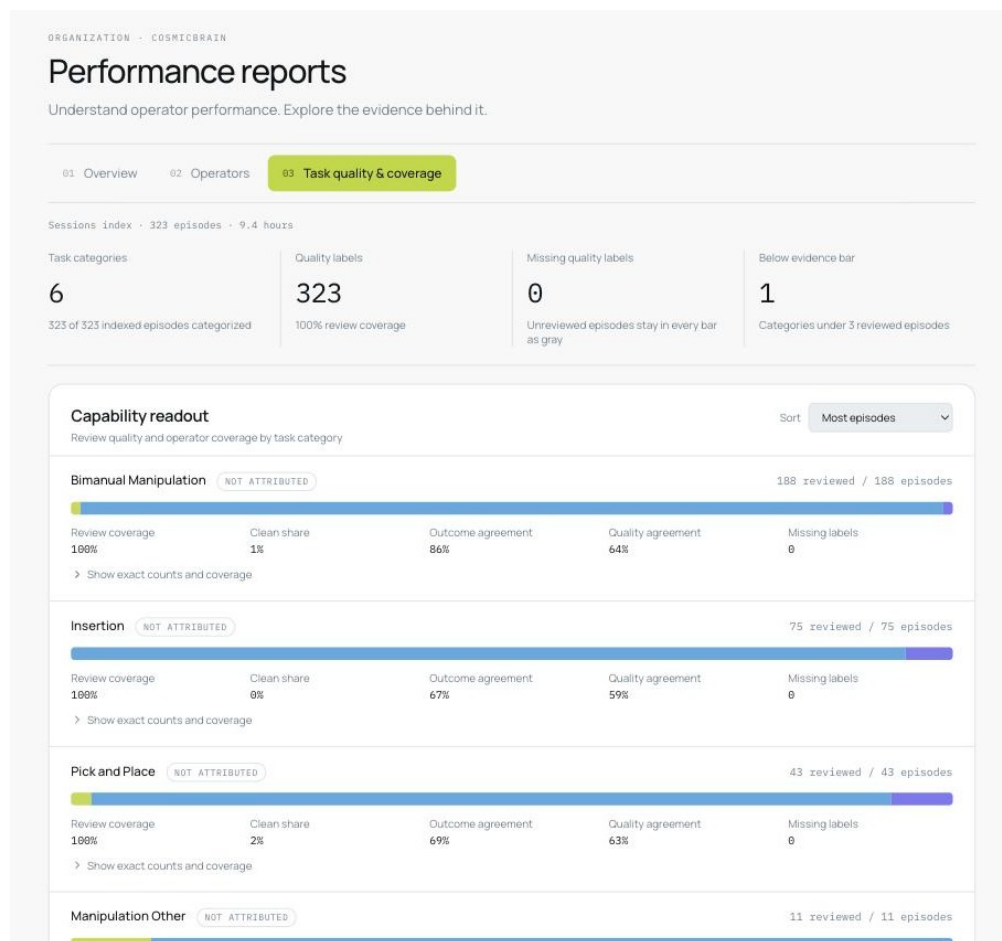


Figure 2: Task-quality report of the audited workspace. Every category is marked *not attributed* because no episode carries an operator identifier.

3.3 Protocol

On 2 October 2026 we read all 323 rows of the episode table (score, outcome label, issue tag, category, duration and task title), the three report views, and the full evidence pages of two episodes selected for contradictory score and label. Task titles were mapped to seven task families by keyword rules (first match wins; families can overlap semantically). Table 1 summarizes the corpus.

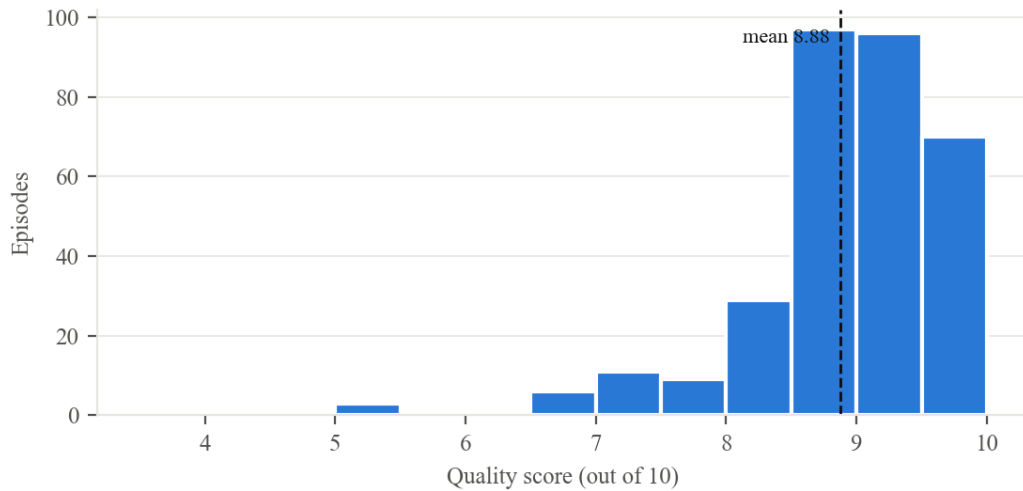
Table 1: Summary of the audited corpus.

Property	Value
Episodes / duration	323 episodes / 9.4 h; mean 105 s (range 20 to 291 s)
Categories (service)	bimanual manipulation 188, insertion 75, pick and place 43, other 11, pouring 5, folding 1
Distinct task titles	189 titles describing roughly 10 underlying tasks
Operator identifiers	none
Review coverage	100% of episodes carry a quality grade

4 Results

4.1 Score and label distributions

Scores are concentrated near the top of the scale (Figure 3, Table 2): 90% of episodes score 8 or higher and 51% score 9 or higher. The outcome distribution reported by the service is 272 success, 39 partial and 12 needs review, an 84.2% success rate (Figure 4). The episode table, however, labels the middle group *inconclusive* (38 episodes) while the report labels it *partial* and lists zero inconclusive episodes, indicating vocabulary drift between views.

**Figure 3:** Distribution of quality scores over 323 episodes. The dashed line marks the mean (8.88).**Table 2:** Episodes by score band.

Score band	Episodes	Share
9.5 to 10.0	70	22%
9.0 to 9.5	96	30%
8.0 to 9.0	126	39%
7.0 to 8.0	20	6%
6.0 to 7.0	6	2%
below 6.0	5	2%

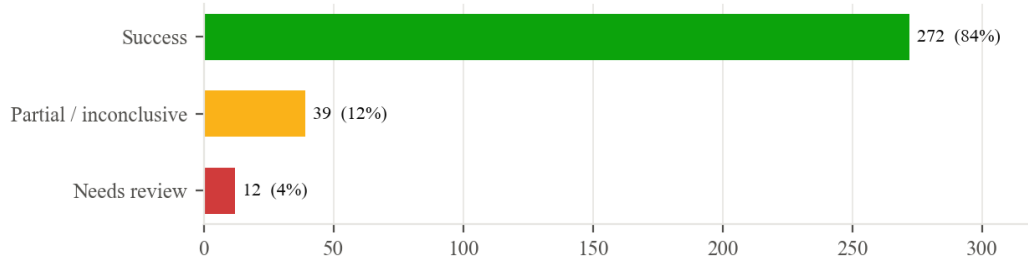


Figure 4: Analysis outcome labels as reported by the service.

4.2 Variation across tasks

Contact-rich and fine-placement tasks score lowest: locker and cloth tasks average 7.41 and medicine-to-first-aid-kit tasks 8.34, while short RAM insertions average 9.30 (Figure 5, Table 3). Insertion episodes are short (31 s on average) and end in a visually distinctive state, which plausibly favors a video-based judge; contact-rich tasks are where a single camera is least informative.

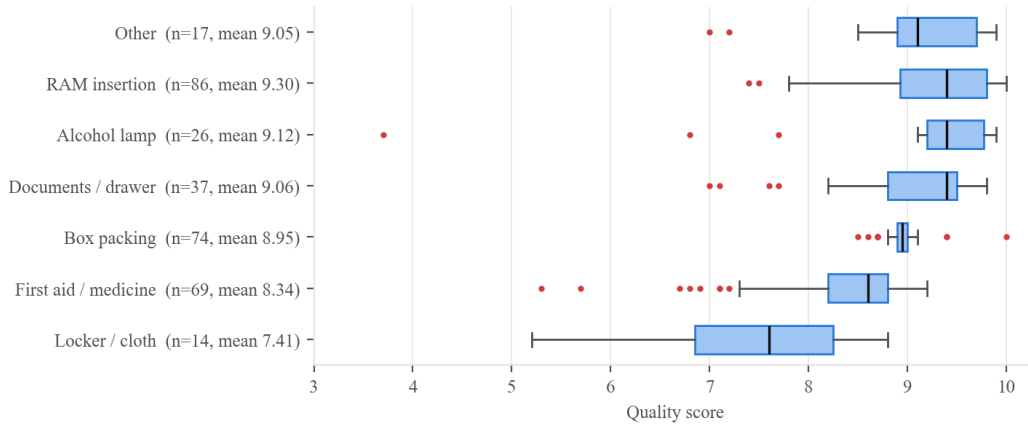


Figure 5: Quality score by task family, sorted by mean. Boxes show the interquartile range; red points are outliers.

Table 3: Score statistics by task family.

Task family	n	Mean	Median	Min	Max	Mean length
RAM insertion	86	9.30	9.4	7.4	10.0	31 s
Alcohol lamp	26	9.12	9.4	3.7	9.9	232 s
Documents / drawer	37	9.06	9.4	7.0	9.8	121 s
Other	17	9.05	9.1	7.0	9.9	126 s
Box packing	74	8.95	8.95	8.5	10.0	158 s
First aid / medicine	69	8.34	8.6	5.3	9.2	80 s
Locker / cloth	14	7.41	7.6	5.2	8.8	106 s

4.3 Reliability of verdicts

The service reports reviewer agreement per category (Figure 6). Outcome agreement ranges from 67% (insertion) to 86% (bimanual), and quality-grade agreement from 59% to 70%. In total 227 of 323 episodes (70%) are flagged for panel disagreement, reaching 81% for insertion. Quality grades are almost uniformly *mixed*: the clean share is 1% for bimanual manipulation (2 of 188), 0% for insertion and 2% for pick and place, which is difficult to reconcile with a score distribution in which 90% of episodes score at least 8.

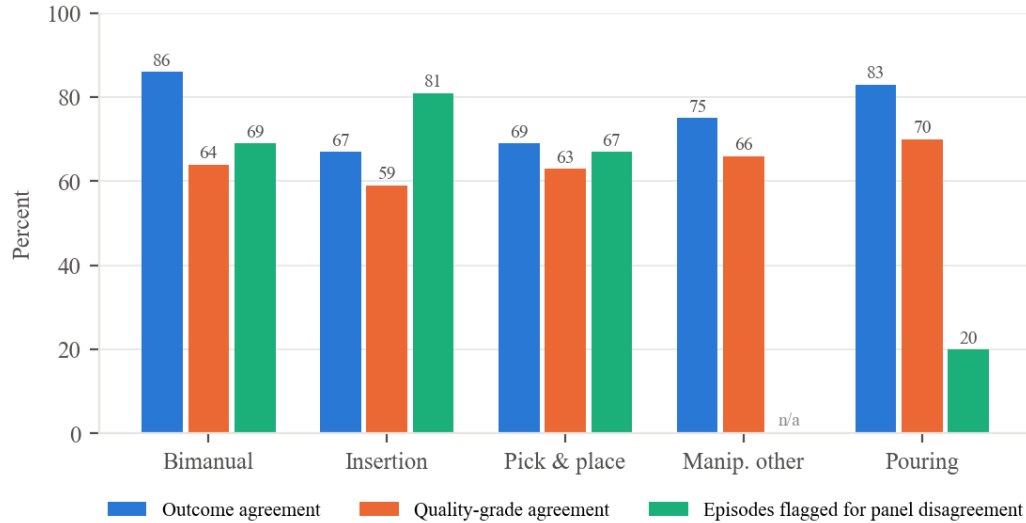


Figure 6: Reviewer agreement and the share of episodes flagged for panel disagreement, by category. The disagreement share for *manipulation other* was not reported in the view we accessed.

Score is also unrelated to execution time. Across all episodes the Pearson correlation between duration and score is 0.07 (Figure 7); the visible clusters correspond to tasks, not to execution quality. A slow, hesitant demonstration and a fast, clean one can therefore receive the same score. Finally, score and label frequently disagree: 6 episodes labeled success score below 7, and 5 non-success episodes score 9 or higher.

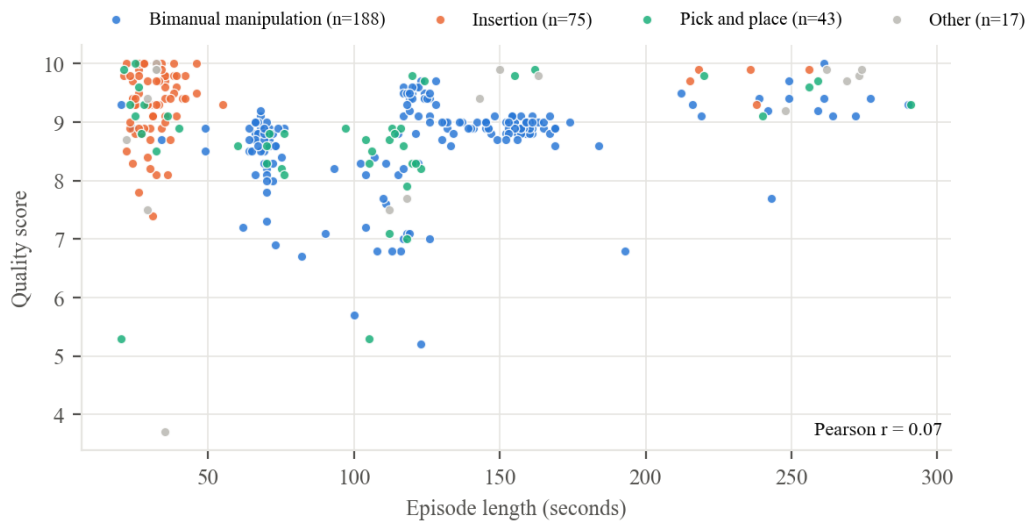


Figure 7: Episode duration against quality score, colored by the service's category (Pearson $r = 0.07$).

4.4 Case analysis

We inspected the evidence pages of the two most contradictory episodes (Figure 8, Table 4). Both state that the score is a *provisional blend of 100% recorded view and 0% robot measurements*, although the dataset includes full joint state and actions.

- **Turn on the desk lamp** (score 3.7, labeled success). The lamp is lit at the end and task completion scores 10. The total is pulled down by an arm-visibility facet of 0, because the scored camera is a wrist view that cannot show the robot's arms. The low score reflects camera selection, not execution.
- **Place the RAM stick into the yellow tray** (score 10.0, labeled needs review). The written evidence states that the insertion attempt is clear but completion is not. Automated completion read 2.0 and was raised to 10 by video review; the service notes that review only examines low readings and can only raise them. Label agreement is 50%, and the task title does not match the observed behavior.

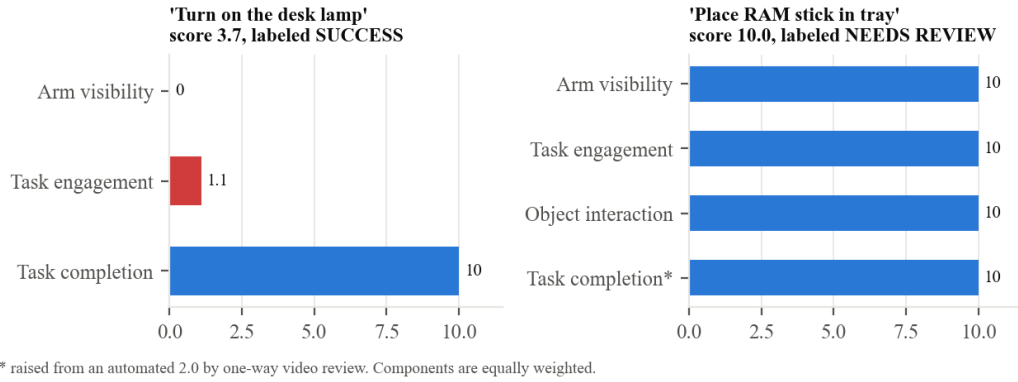


Figure 8: Published facet scores for the two contradictory episodes. Facets are equally weighted within each episode.

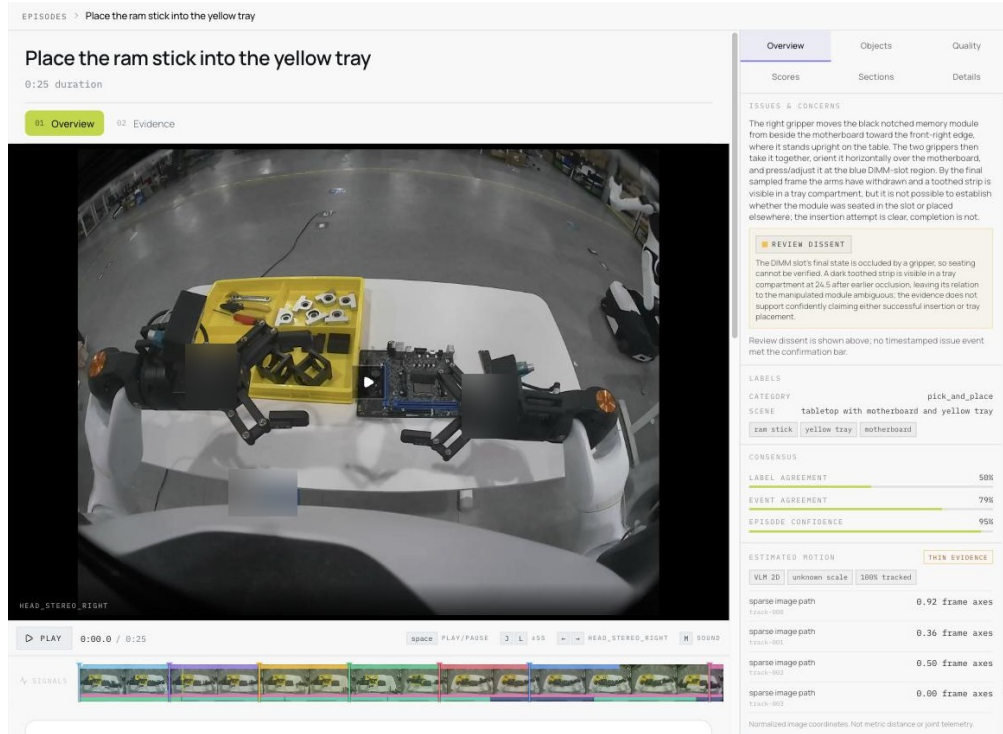


Figure 9: Evidence page for the 10.0 / needs-review episode: one camera, review dissent, 50% label agreement, and motion estimated from 2-D video and marked *thin evidence*.

Table 4: Lowest-scoring and contradictory episodes.

Score	Label	Task	Observation
3.7	success	Turn on the desk lamp	Completion 10; arm visibility 0 from wrist camera
10.0	needs review	Place the RAM stick into the yellow tray	Completion raised 2.0 to 10; label agreement 50%
5.2	needs review	Place the towel on the cabinet door	Lowest-scoring locker task
5.3	needs review	Place the white medicine bottle into the first aid box	Small-object placement
5.3	inconclusive	Place the clothes into the laundry basket	20 s episode
5.7	inconclusive	Place the white medicine bottles into the medical kit	Small-object placement

4.5 Summary of findings

Table 5: Findings of the audit.

ID	Finding	Evidence	Consequence
F1	Video-only scoring	0% weight on robot measurements	Ignores recorded state and actions
F2	Camera-dependent facets	3.7 on a successful episode (wrist view)	Scores track camera choice
F3	One-way correction	Review can only raise low readings	Upward bias; compressed top of scale
F4	Low reviewer agreement	70% flagged; 59 to 70% grade agreement	Verdicts are contested
F5	Score and grade disagree	90% score at least 8; about 1% clean	Unclear which signal to trust
F6	Efficiency ignored	Duration vs score $r = 0.07$	Slow and fast demos rank equally
F7	Label vocabulary drift	partial vs inconclusive across views	Brittle downstream automation
F8	Task-title sprawl	189 titles for about 10 tasks	Fragmented per-task statistics
F9	No operator attribution	Every category not attributed	Operator scoring inactive

Taken together, the scores are useful for *triage*: they surface, within minutes, the 20 to 60 episodes that merit human review, with readable evidence. They are not yet credible enough to filter training data or to certify datasets for third parties, and the decisive test of whether higher-scored episodes yield better policies has not been run.

5 A Telemetry-First Verifier Stack

We propose an evaluation layer in which a single ensemble of verifiers produces one verdict per episode, policy rollout or live task attempt, recorded with its confidence and evidence in a versioned ledger (Figure 9). The same verdicts drive data quality control, policy evaluation and runtime hand-off; takeovers and failures flow back into capture as labeled training data.

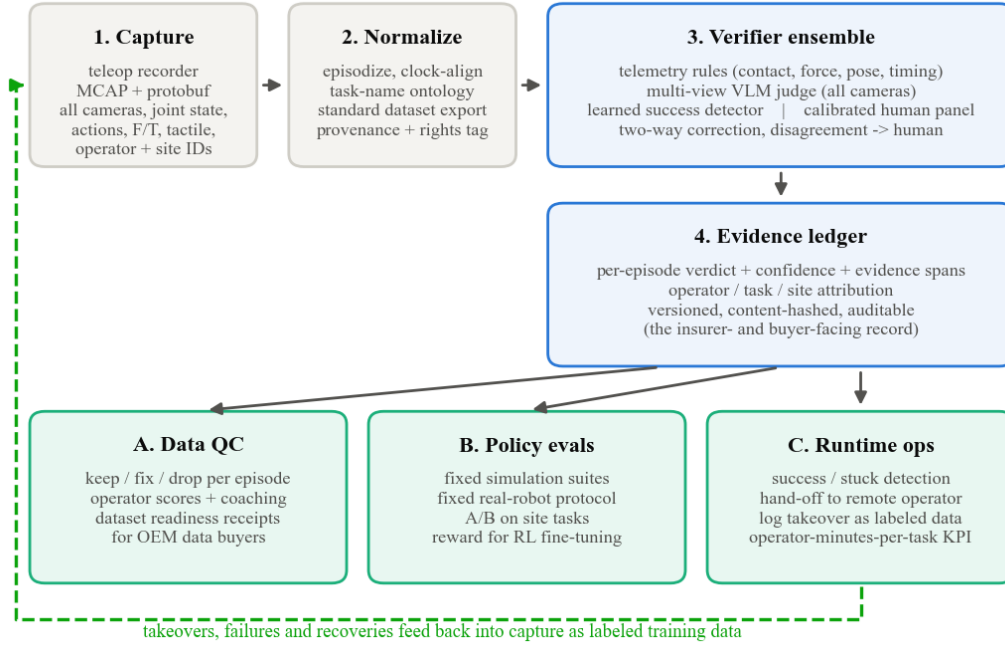


Figure 10: Architecture of the proposed evaluation layer: capture and normalization, a verifier ensemble, an evidence ledger, and three consumers. The dashed path returns takeovers, failures and recoveries to capture as labeled data.

5.1 Design principles

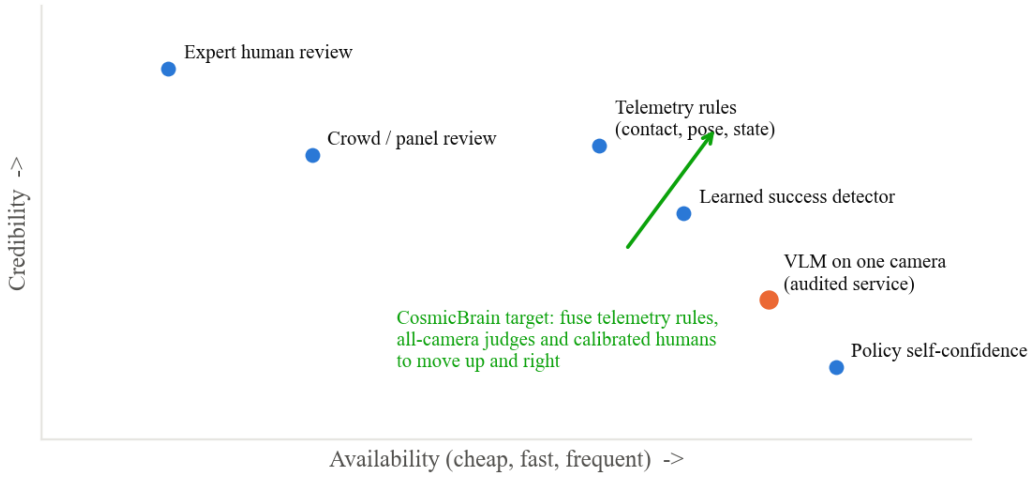
- **Telemetry first.** Joint state, actions, gripper state, force/torque and tactile channels are cheaper and more objective than pixels; video is used where telemetry is silent (addresses F1).
- **All cameras, camera-invariant facets.** No facet may depend on which single camera was selected (F2).
- **Two-way correction.** Humans audit a random sample of high scores as well as low ones, and every correction is logged (F3).
- **Disagreement as a routing signal.** When verifiers disagree, the episode is routed to a calibrated human and the disagreement is stored (F4, F5).
- **Efficiency as a facet.** Time to completion relative to a per-task baseline enters execution quality (F6).
- **Attribution and ontology at capture.** Operator, site, robot, task (from a fixed ontology), outcome vocabulary and data-rights tags are written at recording time (F7 to F9).

5.2 Verifier ensemble

Table 5 lists the verifier families. Their combination aims to move the operating point up and to the right of the credibility-availability frontier (Figure 10).

Table 6: Verifier families in the proposed ensemble.

Verifier	Signal	Detects	Credibility	Cost	Role
Telemetry rules	state, actions, gripper, F/T, timing	drops, collisions, stalls, unsafe velocity	high (in scope)	near zero	hard gates
Multi-view VLM judge	all cameras	semantic success, wrong object or placement	medium	low	outcome and evidence text
Learned success detector	video and telemetry	task-specific success at scale	medium-high	low	runtime hand-off; RL reward
Calibrated human panel	replay and ledger	ambiguous cases; ground truth	high	high	disagreements; audits
Simulation replay	recorded actions	kinematic infeasibility; sync errors	medium	medium	dataset certification

**Figure 11:** Qualitative placement of verifier families on the credibility-availability trade-off described in [1]. Positions are illustrative.

5.3 Metrics

Table 7: Metrics produced by the evaluation layer.

Metric	Definition	Consumer
Episode verdict	success, partial or fail with calibrated confidence and evidence spans	all
Readiness	share of hard gates passed (synchronization, completeness, telemetry present, rights)	data QC; buyer receipts
Execution quality	completion x efficiency (vs task baseline) x smoothness x safety events	operator scoring; ranking
Operator score	difficulty-adjusted execution quality per operator, with trend	coaching; certification
Policy success rate	verified successes over attempts on a fixed suite, with 95% CI	policy evaluation
Operator minutes per task	teleoperation and supervision minutes per verified completed task	operations KPI
Verifier agreement	Cohen's kappa between verifiers and against human audits	health of the stack

6 Validation Protocol

A scoring method should be trusted for training-data selection only if it passes the three tests in Table 7. The first is decisive: a score that does not predict policy performance has no value as a filter, however plausible its evidence text.

Table 8: Validation protocol for scoring methods.

Test	Procedure	Acceptance criterion
Predictive validity	Train the same policy (e.g., ACT or diffusion in LeRobot [4]) on equal-size top-ranked, bottom-ranked and random subsets; evaluate on a fixed simulated and real suite.	top > random > bottom, significant at the 95% level
Human agreement	Expert panel labels a stratified random sample of 100 episodes, including high scorers.	Cohen's kappa at least 0.6; no directional bias
Invariance	Re-score the same episodes from different single cameras and with telemetry withheld.	stable verdicts across cameras; telemetry improves agreement

Policy evaluation follows the same discipline: fixed task suites in simulation [7] and a written real-robot protocol per deployment site (task list, trial counts, reset procedure and success definitions), results reported with confidence intervals, and a shared policy-environment adapter in the style of XPolicyLab [3].

7 Discussion and Limitations

Implications. Verification is the layer through which data quality, policy quality and operational cost are all measured. For an operator of supervised robots, the central business metric, operator minutes per completed task, is only as trustworthy as the verifier that decides a task was completed. This argues for owning the verifier stack while using external services for triage and inspection (Table 8).

Table 9: Build-versus-buy summary.

Option	Strength	Gap	Use
Service A (audited) [8]	operator, task and episode scoring; readable evidence	video-only; one-way correction; no evals	triage once data has operator IDs
Data-readiness service [10]	readiness checks; annotation layers	hosted only; early	monitor
Commercial visualization platform	mature visualization and fleet data	no verdicts or evals	later, fleet operations
Open-source visualization tool	inspection and queries	no verdicts	inspection now
In-house stack	telemetry verifiers, ledger, takeover data	engineering effort	build

Limitations. We audit one dataset and one service version through its web interface; quality-grade counts were available per category rather than per episode; task families are keyword-derived; the trade-off figure is qualitative; and we have not yet trained policies to test predictive validity. Results may not transfer to other robots, tasks or vendors.

Data rights. The audited episodes come from a gated public dataset [9] uploaded to a third-party service. Licence terms for such uploads should be confirmed before further use; subsequent trials should use first-party recordings.

8 Conclusion

Automated, video-only demonstration scoring produced high and compressed scores that disagreed with its own quality grades, depended on camera choice, were corrected only upward, and were contested by its own reviewers. Such scores help locate episodes worth inspecting but should not yet decide what enters a training set. A telemetry-first ensemble with two-way correction, disagreement routing and an auditable ledger, validated against downstream policy performance, offers a path to verdicts that are both available and credible.

References

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- [10] Commercial data-readiness and annotation service (vendor anonymized), 2026.

A Implementation Plan

Figure 11 and Table 9 give a 16-week plan for the components in Section 5.

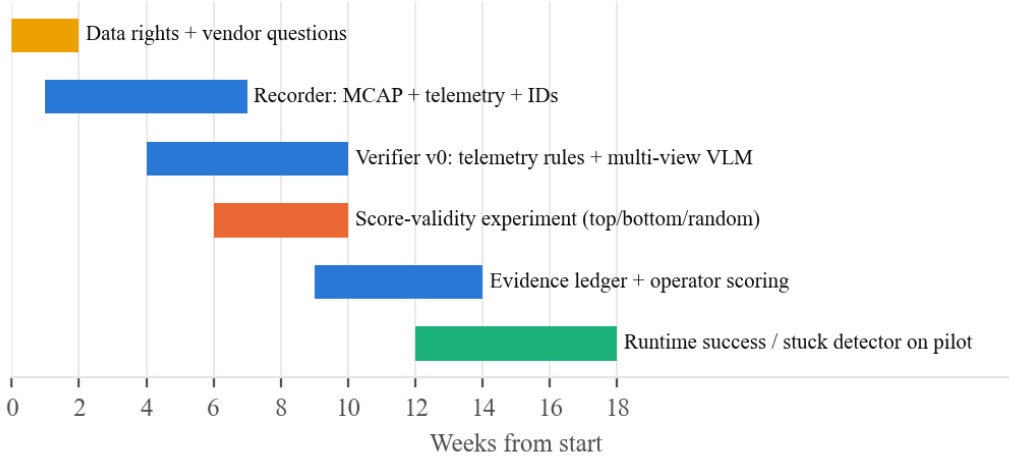


Figure 12: Implementation plan in weeks.

Table 10: Milestones.

Weeks	Deliverable	Exit criterion
0 to 2	Data-rights review; written questions to the vendor	licence position documented
1 to 7	Recorder: MCAP, telemetry, operator, site and task IDs	50 labeled episodes export to LeRobot and replay in simulation
4 to 10	Verifier v0: telemetry rules and all-camera VLM judge	agreement with human audit measured
6 to 10	Predictive-validity experiment (Table 7)	top, bottom and random results with CIs
9 to 14	Evidence ledger and operator scoring	per-operator trends on first-party sessions
12 to 18	Runtime success and stuck detector	used for hand-off on the first pilot

B Capture Schema

Episodes are recorded as MCAP with protobuf messages and exported to LeRobot v3. Each episode contains all camera streams; measured joint state; commanded actions after safety clamps; raw operator XR input; gripper, force/torque and tactile channels where present; safety and watchdog events; takeover events (operator, time, reason code); and metadata for operator, site, robot, task (from a fixed ontology), outcome and data rights. The per-frame action-source flag in [9] is a useful precedent for recording whether a human or a policy was in control.

C Reproducibility

The per-episode table used for all figures (score, duration, category, outcome label, task family and issue flag for 323 episodes) and the scripts that generate every figure and table are stored with this document. Figures use a colorblind-validated categorical palette; green, amber and red are reserved for outcome states.